Anirban Chakrabarti

Areas of Expertise

Applied Cryptography, Privacy-Preserving ML, Blockchain Privacy

Contact Information

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Online Presence

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SUMMARY

PhD in Computer Science (thesis defended Nov 2025) with research and engineering experience across applied cryptography, privacy in distributed systems, and blockchain privacy (encrypted mempools). Currently at Microsoft Research India (since Sep 2025) working on zero-knowledge techniques for LLM inference with a focus on fairness. Proven track record of designing secure protocols and building prototypes in Rust/C++/Python, with 6+ years of industry experience delivering and supporting large-scale systems handling sensitive data.

CORE SKILLS

- Cryptography: threshold encryption, traitor tracing, accountability
- Zero-knowledge proofs: protocol design, efficiency trade-offs
- Blockchain privacy: mempool privacy, MEV-aware protocol design
- Secure cloud storage: deduplication with privacy guarantees
- Programming: Rust, C++, Python, Java, Shell
- Systems: Linux/Unix, distributed systems basics
- Engineering: prototyping, benchmarking, reproducibility
- Communication: academic writing, collaboration with cross-functional teams

EXPERIENCE

Microsoft Research India — Research Intern (Applied Cryptography & Secure AI)
Sep 2025 – Present

- Ongoing research on zero-knowledge proof techniques for LLM inference with an emphasis on fairness and verifiability.
- Translate cryptographic constructions into implementable prototypes and assess feasibility for real-world ML inference settings.
- Collaborate with research mentors and peers to refine problem formulation, assumptions, and evaluation methodology.

Technical University of Darmstadt, Germany — Research Intern (Hosted by Prof. Sebastian Faust)
May 2024 – Oct 2024

- Designed threshold cryptographic protocols to distribute trust among multiple parties to protect sensitive information.
- Developed protocol ideas to reduce Miner Extractable Value (MEV) misuse in decentralized finance (DeFi) scenarios.

Cognizant Technology Solutions, Kolkata and Hongkong — Product Specialist (Technical)
Apr 2012 – Aug 2018

- Worked on the platform migration of a comprehensive mandatory provident fund application from mainframe systems to AIX, including developing software solutions to facilitate the migration and ensure seamless functionality in the new environment.

- Led offshore delivery and automation initiatives and contributed to production support for applications managing sensitive client data by ensuring compliance with regulatory standards.
- Worked across Unix/Linux, Windows Server, and databases (Oracle/DB2) in regulated environments.
- Developed a product in Unix shell script that functioned as an equivalent to IBM Job Control Language (JCL), enabling complex batch job management and automation for enterprise workflows.

EDUCATION

Indian Institute of Science (IISc), Bengaluru — PhD in Computer Science
Aug 2019 – Nov 2025

- Thesis: "Enhancing Privacy and Efficiency of Distributed Encryption"
- Advisor: Prof. Bhavana Kanukurthi

MCKV Institute of Engineering (WBUT), India — B.Tech. in Information Technology 2007 – 2011

SELECTED PUBLICATIONS & RESEARCH HIGHLIGHTS

Secure Fuzzy Deduplication: Definitions and Constructions
A. Chakrabarti, S. Gupta, B. Kanukurthi, G. Shankar.
Proceedings of the 2025 Cloud Computing Security Workshop, pp. 80–94.

- Developed a secure deduplication system designed to remove redundant data while ensuring that unique content is preserved within cloud storage environments.
- Implemented practical system constructions using programming languages such as C++ and Python, alongside Unix shell scripting to facilitate robust system functionality.
- Demonstrated the effectiveness of the deduplication system by validating its performance on real-world datasets, including images and genomic sequences.

Silent Threshold Traitor Tracing & Enhancing Mempool Privacy
A. Chakrabarti, M. Maitra, A. Mondal, K. Sehgal.
Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security, pp. 1769–1783.

- Introduced an innovative silent threshold traitor tracing scheme aimed at strengthening privacy in applications such as encrypted Mempools, private voting systems, and sealed-bid auctions.
- Implemented and benchmarked prototype using Go programming language

Traceable Threshold Batch Encryption with Applications to Enhancing Mempool Privacy
A. Chakrabarti, M. Maitra, A. Mondal
Manuscript under submission at ACM ASIACCS 2026.

- A prototype was developed in Rust, leveraging GitHub Copilot to automate code generation and minimize the need for handwritten coding. This approach significantly accelerated development and ensured consistency in implementation.

TEACHING & AWARDS

- Teaching Assistant: Theoretical Foundations of Cryptography (IISc) - Prepared assignments and supported student problem-solving.
- GATE 2019 (CS): Score 774, AIR 330.
- Professional Recognition (Cognizant, Sep 2017): Recognized for problem-solving and cross-team collaboration.

PROFESSIONAL SERVICE

- Conference Reviewer: Participated as a reviewer for ASIACRYPT 2024, ACNS 2026, and CRYPTO 2026.